

CHEMICAL EQUILIBRIUM

16-12-24

L-01

CHEMICAL REACTION :

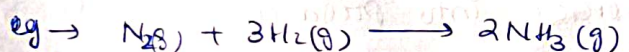
- Symbolic representation of any chemical change in terms of reactants and products is called chemical reaction.

TYPES OF CHEMICAL REACTION :

① On the basis of physical state.

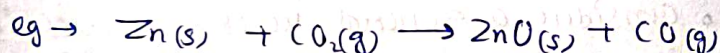
@ Homogenous reaction :

- all reactants and products are in same phase.



② Heterogenous reaction :

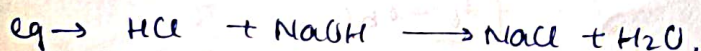
- Reactants and products are in two phases.



② On the basis of speed :

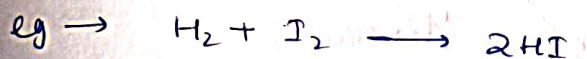
@ Fast reactions :

- generally these reactions are ionic reactions.



③ Slow reaction :

- generally these reactions are molecular reactions.

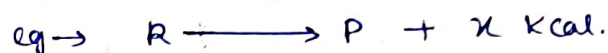


CHEMICAL EQUILIBRIUM

③ On the basis of heat :

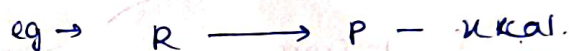
@ Exothermic reaction :

- Heat is evolved in these type of chemical reactions.



④ Endothermic reaction :

- Heat is absorbed in these type of chemical reaction.



④ On the basis of direction :

@ Irreversible reaction

- chemical reaction in which products cannot be converted into reactants.



initially, $[\text{A}] \neq [\text{B}] \Rightarrow \text{high}$

$[\text{C}] \neq [\text{D}] \Rightarrow \text{Low}$

with time concn of reactant ↓
concn of product ↑

after complete reaction,

$$[\text{reactant}] \approx 0$$

- Acid-base neutralisation reaction.



- Precipitation reaction.



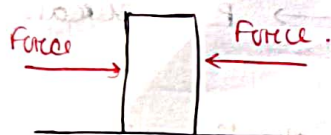
Combustion reaction :



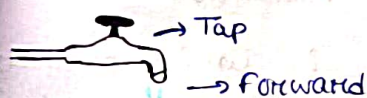
- generally irreversible reaction proceeds only in forward direction.
- generally possible in open container.
- These type of reaction can never attain equilibrium.

(B) Reversible reaction :

- Chemical reactions in which products can be converted back to reactants is called reversible process.



Static equilibrium.



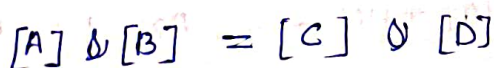
dynamic equilibrium
 Rate of Forward = Rate of backward process



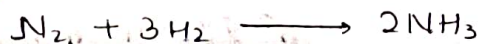
Initially $[A] \text{ \& } [B] \Rightarrow \uparrow$

After sometime,
 $[C] \text{ \& } [D] \Rightarrow \uparrow$ & there will form A & B.

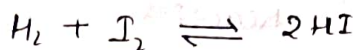
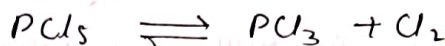
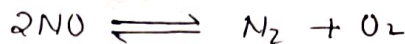
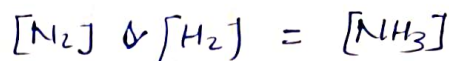
At equilibrium,



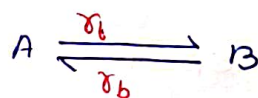
eg $\rightarrow$



Rate of forward = rate of backward.



- Reversible reaction proceed in forward as well as backward direction.
- Possible in closed container.
- There can attain equilibrium
- Reactants are never completely converted into product.
- The most important characteristic property of the reversible reaction is that - it always attains a state of chemical equilibrium.
- Consider a general reversible reaction in a closed vessel. $\rightarrow$



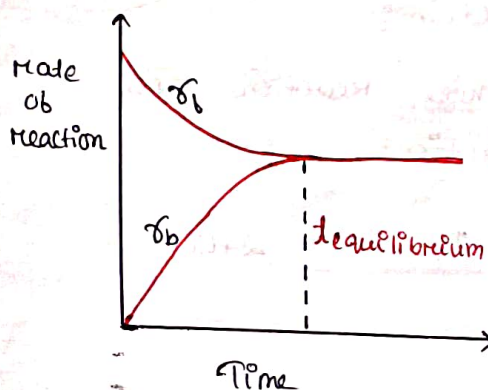
$r_f \rightarrow$ Rate of forward rxn

$r_b \rightarrow$ Rate of backward rxn.

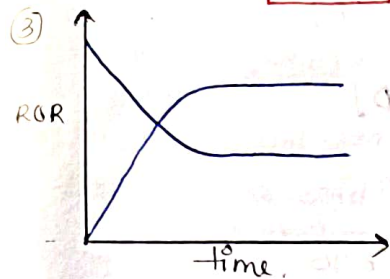
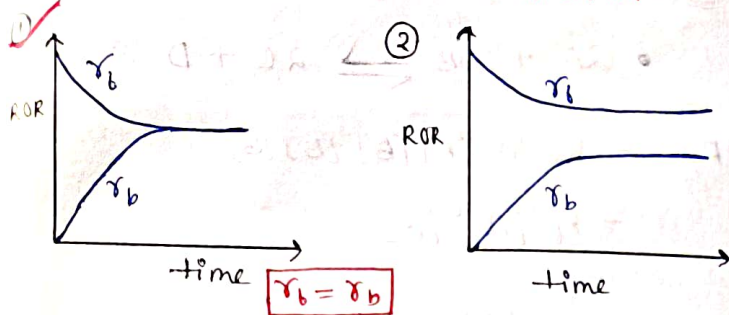
at equilibrium,

$$r_f = r_b$$

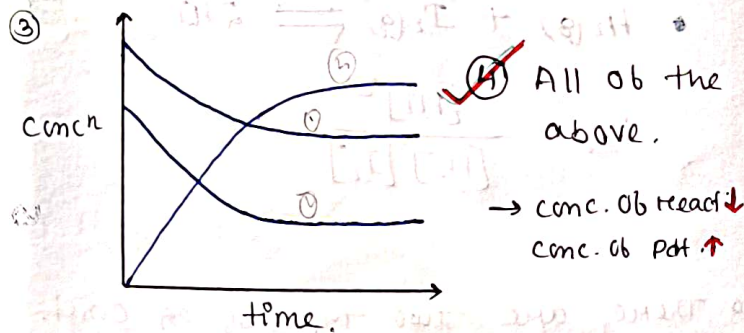
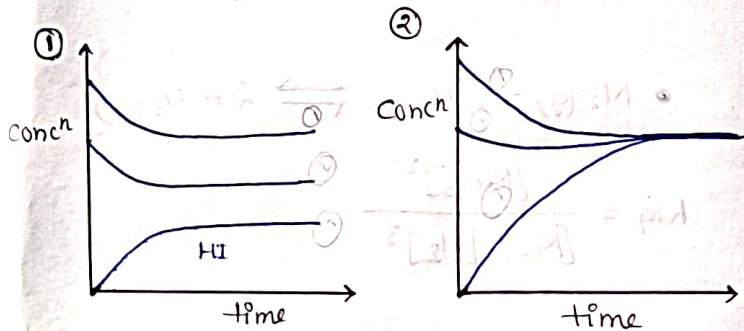
Graph :



8 which is the correct graph.



9 Which one is correct?



CHARACTERISTICS OF EQUILIBRIUM:

① The concn. of reactants and products do not change with time.

② Chemical equilibrium is dynamic in nature i.e., the reaction although appears to be stopped but actually takes place in both the direction with the same speed.

③ Chemical equilibrium can be approached from both sides.

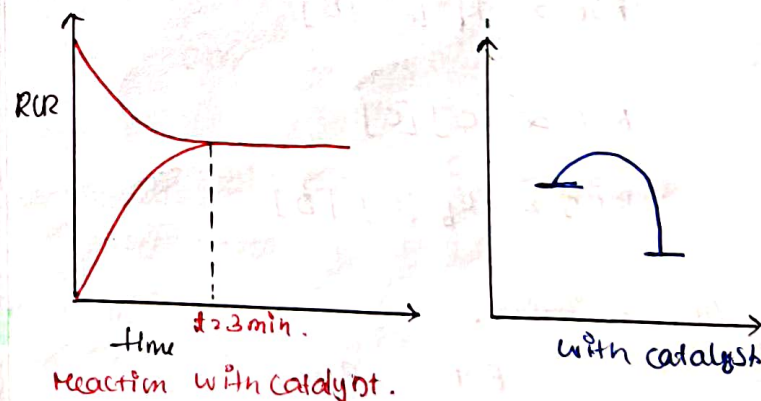
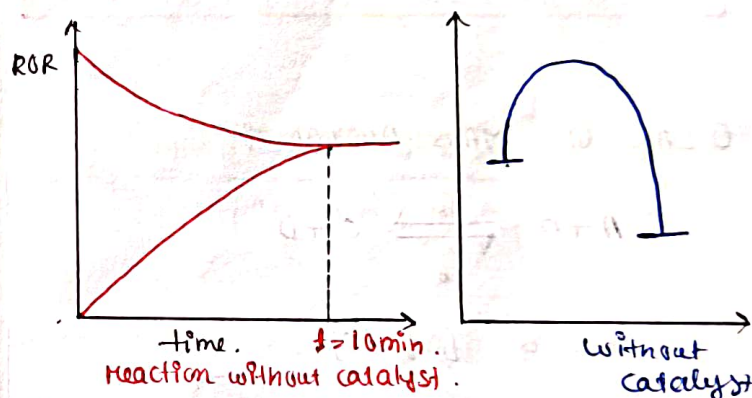
at equm.

each reactant and product has a const. concn and this is independent of the fact whether the reaction starts from forward direction or backward direction with the reactant or with the product.

of the system

④ The measurable properties like temp, concn, colour, density do not undergo any change with time at the chemical equilibrium conditions.

⑤ Equilibrium is not affected by the presence of catalyst. it will only help to attain the equilibrium more rapidly.



⑥ There are two types of equilibrium.

(a) Homogeneous equilibrium

(b) Heterogeneous equilibrium.

● ACTIVE MASS :

o The term active mass means the concn of reactants expressed in moles per litre (molar concn).

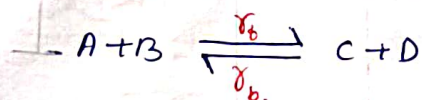
OR.

The pressure in reacting gas in equilibrium. In case of gases and solutions the molar concn means the no. of gram molecules present per litre.

o Active mass is usually expressed by enclosing the symbol of the reactant in 3rd sq. bracket.

$$\text{Active mass} = \frac{\text{no. of gm. molecules of subs.}}{\text{Volume in litre.}}$$

● LAW OF MASS ACTION :



$$R_{OF} \propto [A][B]$$

$$R_{OF} = k_f [A][B]$$

$$R_{OB} \propto [C][D]$$

$$R_{OB} = k_b [C][D]$$

At equilibrium,

$$R_{OF} = R_{OB}$$

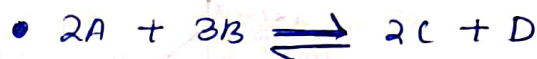
$$k_f [A][B] = k_b [A][B]$$

$$K_{eq} = \frac{k_f}{k_b} = \frac{[C][D]}{[A][B]} = \frac{[P]}{[R]}$$

o Stability of reactant rises when K ↓

o Stability of product rises when K ↑

Q Find the K_{eq} .



$$R_{OF} = k_f [A][A][B][B][B]$$

$$= k_f [A]^2 [B]^3$$

$$R_{OB} = k_b [C][C][D]$$

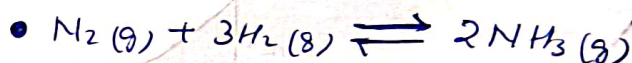
$$= k_b [C]^2 [D]$$

At equilibrium,

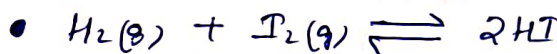
$$R_{OF} = R_{OB}$$

$$k_f [A]^2 [B]^3 = k_b [C]^2 [D]$$

$$K_{eq} = \frac{k_f}{k_b} = \frac{[C]^2 [D]}{[A]^2 [B]^3}$$



$$K_{eq} = \frac{[NH_3]^2}{[N_2][H_2]^3}$$



$$K_{eq} = \frac{[HI]^2}{[H_2][I_2]}$$

o There are two types of eq. const.

K_{eq} .

K_c

eq. const. in terms of concn

K_p

only for gas.

eq. const. in terms of concn.

$$[] = \text{concn} \Rightarrow \frac{n}{V}$$

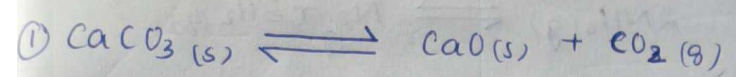
$$[] \Rightarrow \text{Pressure of particular gas.}$$

$$PV = nRT$$

$$P = \frac{n}{V} RT$$

$$P \propto \frac{n}{V}$$

8 Write expression and units for K_p and K_c for the following rxns.



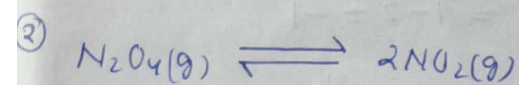
$$K_c = \frac{[\text{CO}_2] [\text{CaO}]^1}{[\text{CaCO}_3]^1} \quad K_p = (P_{\text{CO}_2})^1$$

$$\text{Unit} = (\text{atm})^1$$

$$K_c = [\text{CO}_2]^1 \quad \text{Unit} = \text{mol/Lit.}$$

→ If solid and gas are present in eq. then conc of solid is equals to 1.

∴ Rule no. 1



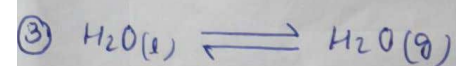
$$K_c = \frac{[\text{NO}_2]^2}{[\text{N}_2\text{O}_4]} \quad K_p = \frac{(P_{\text{NO}_2})^2}{(P_{\text{N}_2\text{O}_4})}$$

$$\text{Unit} = \frac{(\text{mol/Lit})^2}{(\text{mol/Lit})}$$

$$\text{Unit} = \text{atm.}$$

$$= \text{mol/Lit.}$$

→



$$K_c = \frac{[\text{H}_2\text{O}]_g}{[\text{H}_2\text{O}]_l} \quad K_p = (P_{\text{H}_2\text{O}})^1$$

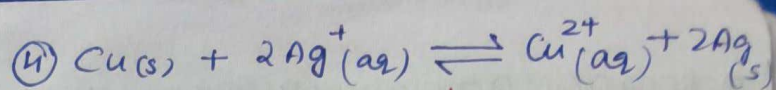
$$\text{Unit} = (\text{atm})^1$$

$$K_c = [\text{H}_2\text{O}]^1$$

$$\text{Unit} = \text{mol/Lit.}$$

→ If liq. and gas are present in equilibrium. then conc of liq equals to 1.

∴ Rule no. 2.



$$K_c = \frac{[\text{Cu}^{2+}]_{aq} [\text{Ag}]_{(s)}^2}{[\text{Cu}]_{(s)}^1 [\text{Ag}^+]_{aq}^2}$$

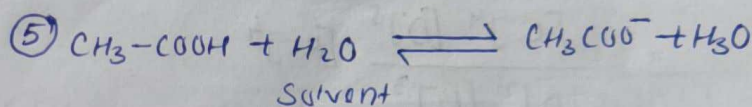
$K_p = \text{undefined}$

$$= \frac{[\text{Cu}^{2+}]_{aq}}{[\text{Ag}^{2+}]^2}$$

$$\text{Unit} = \frac{\text{mol/Lit}}{(\text{mol/Lit})^2}$$

$$= \text{Lit/mol.}$$

→ when solid and liq are present in eq. then conc of solid is equals to 1.



$$K_c = \frac{[\text{CH}_3\text{COO}^-] [\text{H}_3\text{O}^+]}{[\text{CH}_3\text{COOH}] [\text{H}_2\text{O}]^1}$$

$$\text{Unit} \rightarrow \text{mol/Lit.}$$

→ when H_2O is behave as solvent. then conc of H_2O is equals to 1.

9 Active mass of two mole of NaCl kept in 4L vessel at NTP is →

✓ ① 1 ② 2 ③ $\frac{1}{2}$ ④ not defined.

Soln:- NaCl → solid

$$[\text{NaCl}] \rightarrow 1$$

RELATION BETWEEN K_p AND K_c



$$K_c = \frac{[C]^4 [D]^5}{[A]^2 [B]^3}$$

$$PV = nRT$$

$$P = \frac{n}{V} RT$$

$$K_p = \frac{(P_C)^4 (P_D)^5}{(P_A)^2 (P_B)^3}$$

$$P = CRT$$

$$K_p = \frac{[C]^4 [D]^5 [RT]^4 [RT]^5}{[A]^2 [B]^3 [RT]^2 [RT]^3}$$

$$= \frac{[C]^4 [D]^5 [RT]^{4+5-2-3}}{[A]^2 [B]^3}$$

$$= \frac{[C]^4 [D]^5}{[A]^2 [B]^3} \cdot [RT]^4$$

$$= \frac{[C]^4 [D]^5}{[A]^2 [B]^3} [RT]^{\Delta n_g}$$

$$K_p = K_c (RT)^{\Delta n_g} \quad K_p = K_c (RT)^{\Delta n_g}$$

$$K_p \rightarrow \text{atm}$$

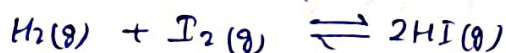
$$K_c \rightarrow \text{mol/Lit.}$$

$$R \rightarrow \frac{1}{12} \text{ OR } 0.082$$

$$T \rightarrow K.$$

$\Delta n_g \rightarrow$ no of gaseous molecules in Pdt
- no of gaseous molecules in reactant.

CASE-I: $\Delta n_g = 0$

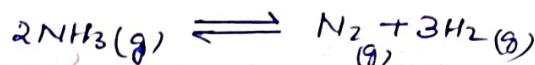


$$\Delta n_g = 2 - 2 = 0$$

$$K_p = K_c (RT)^{\Delta n_g}$$

$$K_p = K_c$$

CASE-II: $\Delta n_g > 1$

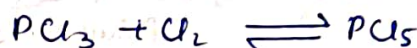


$$\Delta n_g = 4 - 2 = +2$$

$$K_p = K_c (RT)^{\Delta n_g}$$

$$K_p > K_c$$

CASE-III: $\Delta n_g < 1$



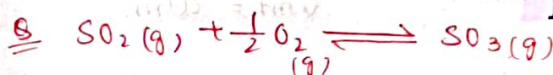
$$\Delta n_g = 1 - 2 = -1$$

$$K_p = K_c (RT)^{\Delta n_g}$$

$$K_p < K_c$$

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L-02



if $K_p = K_c (RT)^{\Delta n_g}$. then find the value of n .

$$\text{Soln: } \Delta n = 1 - 1 - \frac{1}{2} = -\frac{1}{2}$$



if $K_p = 1.44 \times 10^{-5}$ at $500^\circ C$ - find the value to K_c .

$$\text{Soln: } K_p = K_c (RT)^{\Delta n_g}$$

$$1.44 \times 10^{-5} = K_c \left(\frac{1}{12} \times 773 \right)^{-2}$$

$$= K_c \left(\frac{1}{12} \times \frac{1}{773} \right)^2$$

$$1.44 \times 10^{-5} = K_c \cdot 4 / 773 \times 773$$

$$K_c = \frac{1.44 \times 10^{-5} \times 773 \times 773}{144}$$

$$= \frac{1.44 \times 10^{-5}}{(0.082 \times 773)^{-2}}$$

● FACTORS AFFECTING EQ. CONST. :

① Temperature :

→ The value of eq. const. changes with the change of temp. If K_1 and K_2 be the eq. const. of a reaction at absolute temp T_1 and T_2 and ΔH is the change in enthalpy then,

$$\log\left(\frac{K_2}{K_1}\right) = \frac{\Delta H^\circ}{2.303R} \left[\frac{1}{T_1} - \frac{1}{T_2} \right]$$

$$\log \frac{K_2}{K_1} = \frac{\Delta H^\circ}{2.303R} \frac{T_2 - T_1}{T_1 T_2}$$

$$\begin{aligned} T_1 &\rightarrow K_1 \\ T_2 &\rightarrow K_2 \\ T_2 &> T_1 \end{aligned} \quad \text{If } T_2 > T_1 \quad \left[\frac{T_2 - T_1}{T_1 T_2} \right] > 0$$

Case-1 :

for any reaction, $\Delta H^\circ = 0$

$$\log K_2 - \log K_1 = 0$$

$$\log K_2 = \log K_1$$

$$K_2 = K_1$$

• with increase in temp there will be no effect on eq. const.

Case-2 :

Exothermic reaction,

$$\Delta H < 0 \rightarrow -ve$$

$$\log K_2 - \log K_1 = -ve$$

$$\log K_2 - \log K_1 < 0$$

$$\log K_2 < \log K_1$$

$$K_1 > K_2$$

• with increase temp. eq. const. decreases.

Case-3 :

Endothermic reaction.

$$\Delta H > 0 \rightarrow +ve.$$

$$\log K_2 - \log K_1 > 0$$

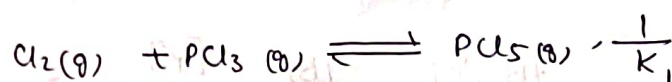
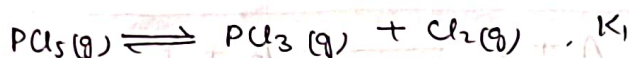
$$\log K_2 > \log K_1$$

$$K_2 > K_1$$

• with increase in temp eq. const. increase.

② The mode of representation of the reaction

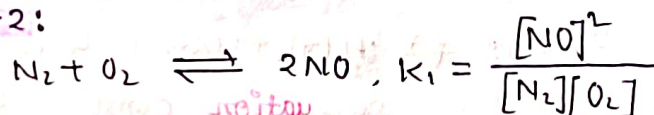
Case-1 :



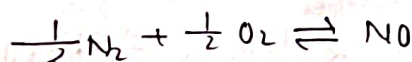
$$K_1 = \frac{[PCl_5]}{[PCl_3][Cl_2]} \quad K_2 = \frac{[PCl_5]}{[PCl_3][Cl_2]}$$

$$K_2 = \frac{1}{K_1}$$

Case-2 :



Divided by 2 in both side.

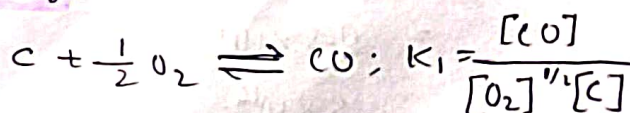


$$K_2 = \frac{[NO]}{[N_2]^{1/2}[O_2]^{1/2}}$$

$$K_2 = (K_1)^{1/2}$$

$$\text{Divide} = K \quad \frac{1}{\text{eq.}}$$

Case-3 :



Multiplied by 2 in both side.



$$K_2 = \frac{[CO]^2}{[C]^2[O_2]}$$

$$K_2 = K_1^2$$

Stoichiometric coeff. $\Rightarrow$ multiply
 $\downarrow$
power.

Case-4:

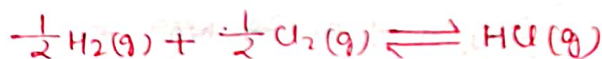


$$K = \frac{[B][Y]}{[A][X]} = \frac{[B]}{[A]} \cdot \frac{[Y]}{[X]}$$

$$K = K_1 \cdot K_2$$

Q-3
The eq. const. (K_c) for the reaction.

$2HCl(g) \rightleftharpoons H_2(g) + Cl_2(g)$ is 4×10^{34} at $25^\circ C$. What is the eq. const. for the reaction —



Soln:- $K' = \left(\frac{1}{K}\right)^{1/2}$

$$= \left(\frac{1}{4 \times 10^{34}}\right)^{1/2}$$

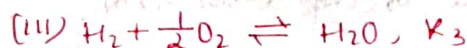
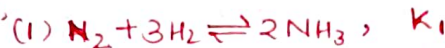
$$= \left(\frac{10^{34}}{4}\right)^{1/2}$$

$$= \frac{1}{2} \times 10^{17}$$

$$= 0.5 \times 10^{17}$$

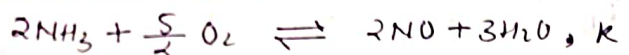
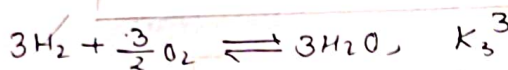
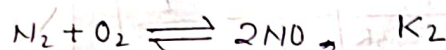
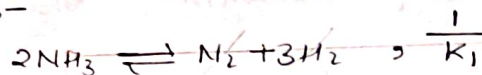
$$= 5 \times 10^{16}$$

Q-3 Consider the following gaseous eqn given below,



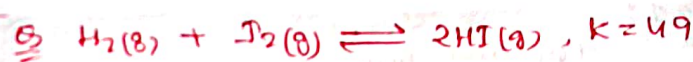
The equilibrium constant for the reaction $2NH_3 + \frac{5}{2}O_2 \rightleftharpoons 2NO + 3H_2O$ in terms of K_1, K_2 and K_3 be.

Soln:-



$$K_{eq} = \frac{1}{K_1} \cdot K_2 \cdot K_3^3$$

$$= \frac{K_2 K_3^3}{K_1}$$



Soln: $K_{eq} = \left(\frac{1}{49}\right)^{1/2}$
 $= \frac{1}{7}$

Mw.
BB-1, 2, 3

19-12-24

L-03

① FACTORS NOT AFFECTING EQ. CONST.:

① concn of reactants and products

② Pressⁿ

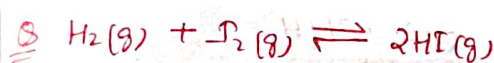
③ Vol^m

④ Presence of inert materials

⑤ The presence of catalyst.

NOTE:

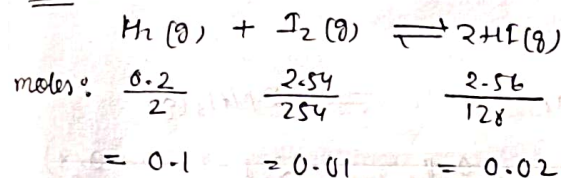
The value of eq. const. depend only on temp^r.



0.2 g H_2 , 2.54 I_2 , 2.56 g HI are obtained at eq. in a 1 L vessel. calculate K_p , K_c for the reaction.

At. mass, $H = 1$, $I = 127$.

Soln:-



conc'n $\frac{n}{V}$ = $\frac{0.1}{1}$ $\frac{0.01}{1}$ $\frac{0.02}{1}$

$= 0.1$ 0.01 0.02

$$K_c = \frac{[HI]^2}{[H_2][I_2]} = \frac{[0.02]^2}{[0.1][0.01]}$$

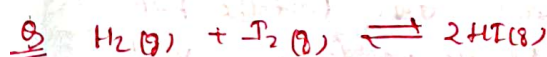
$$= \frac{0.02 \times 0.02}{0.1 \times 0.01}$$

$$= \frac{0.04}{0.1} = 0.4$$

$$K_p = K_c (RT)^{\Delta n_g}$$

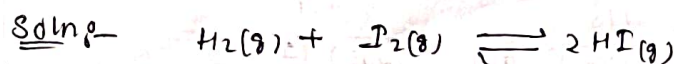
$$= K_c (RT)^0$$

$$K_p = K_c = 0.4$$



3 mole of H_2 and 3 mole I_2 are kept in a 10L container. If K_c for the rxn at $25^\circ C$ is 49. Find the no. of moles of I_2 at eq. at $25^\circ C$.

- ① 0.33 ② 1.33 ③ 0.66 ④ 1.66



initially: 3mole 3mole 0

(mole/l)

at eq: $(3-x)$ $(3-x)$ $2x$

conc'n $\left(\frac{n}{V}\right)$: $\frac{3-x}{10}$ $\frac{3-x}{10}$ $\frac{2x}{10}$

$$K_c = \frac{[HI]^2}{[I_2][H_2]} = \frac{[2x/10]^2}{\left[\frac{3-x}{10}\right]\left[\frac{3-x}{10}\right]}$$

$$= \frac{[2x]^2}{[3-x]^2} = \left[\frac{2x}{3-x}\right]^2$$

$$49 = \left(\frac{2x}{3-x}\right)^2$$

$$7 = \frac{2x}{3-x}$$

$$21 - 7x = 2x$$

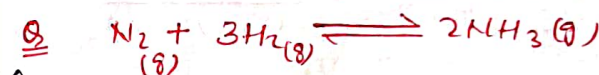
$$21 = 9x$$

$$x = \frac{21}{9} = 2.33$$

no. of moles of I_2 at eq. = $3 - x$

$$= 3 - 2.33$$

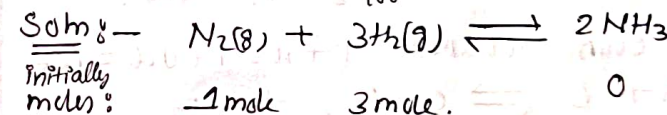
$$= 0.66$$



1 mole of N_2 and 3 moles H_2 are made to react in a 4L vessel if 25% H_2 converts to NH_3 find value of K_c .

① $\frac{2^7}{3^7}$ ② $\frac{2^{10}}{3^7}$ ③ $\frac{2^7}{3^{10}}$ ④ $\frac{2^{10}}{3^{10}}$

$n = 0.25 \left(\frac{25}{100}\right)$ $3 \times \frac{25}{100} = 0.75$ $4 \times \frac{0.7}{25} = 0.12$



finally: $(1-x)$ $(3-3x)$ $2x$

$$(1-0.25) \quad (3-0.75) \quad (2 \times 0.25)$$

$\frac{n}{V} = \frac{1-0.25}{4}$ $\frac{3-0.75}{4}$ $\frac{2 \times 0.25}{4}$

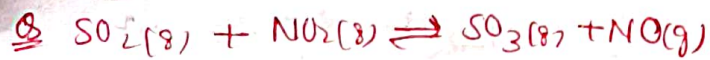
$$K_c = \frac{[0.5]^2}{\left[\frac{2.25}{4}\right]\left[\frac{0.75}{4}\right]}$$

Shanti se kar answer da jayga

$$= \frac{0.5 \times 0.5}{2.25 \times 0.75}$$

$$= \frac{0.25}{1.6875}$$

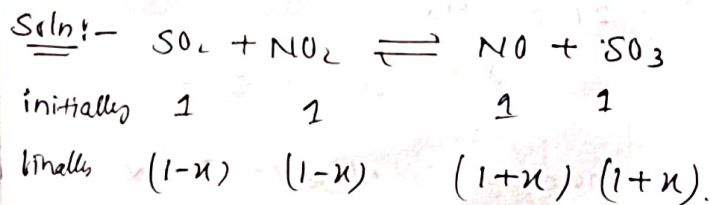
$$= \frac{2^{10}}{3^7}$$



The eq. const for this reaction is 16. If 1 mole of each of 4 gases is enclosed in a vessel of volume 1L. find the eq. const. find the eq. const. of NO.

① 0.4 mole/lit. ② 0.6 mole/lit. ③ 1.4 mole/lit.

④ 1.6 mole/lit.



Concⁿ (M) = $\frac{1-x}{1}$ $\frac{1-x}{1}$ $\frac{1+x}{1}$ $\frac{1+x}{1}$

$K_c = \frac{[SO_3][NO]}{[SO_2][NO_2]} = \frac{(1+x)(1+x)}{(1-x)(1-x)}$

16 = $\frac{(1+x)^2}{(1-x)^2}$

$x = \frac{1+x}{1-x}$

4 - 4x = 1 + x

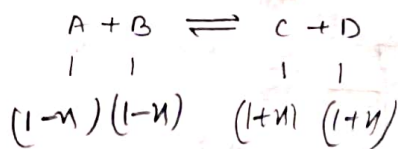
5x = 3

x = 3/5 = 0.6

eq. const. of NO = 1 + x = 1 + 0.6 = 1.6

Q A + B ⇌ C + D. The eq. const. of this reaction is 100. If initially 4 mole each of 4 species is taken then find concn of D.

① 0.818 ② 1.818 ③ 0.182 ④ 1.182



100 = $\frac{(1+x)^2}{(1-x)^2}$

10 = $\frac{1+x}{1-x}$

10 - 10x = 1 + x

11x = 9

x = 9/11

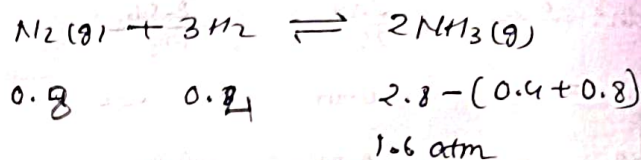
[D] = 1 + x = 1 + 9/11 = 20/11 = 1.818



If partial pressure of N₂ and H₂ are 0.8 atm and 0.4 atm respectively at eq. and total pressure is 2.8 atm find K_p.

① 25 ② 50 ③ 75 ④ 100

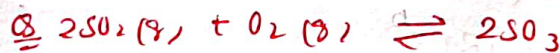
Soln:-



$K_p = \frac{(P_{NH_3})^2}{(P_{N_2})(P_{H_2})^3}$

= $\frac{5^2 \times 10}{0.8 \times 0.4 \times 0.4 \times 0.4}$

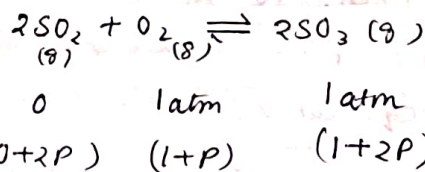
= 50



SO₂ and O₂ gas mixed in a container having initial pressure 4 atm each. If K_p = 1.87 atm⁻¹ then find the partial pressure of SO₂ at eq. (at const volⁿ).

① 0.2 ② 0.8 ③ 0.6 ④ 0.4

Soln:-



$K_p = \frac{[SO_3]^2}{[SO_2]^2[O_2]} = \frac{(1+2P)^2}{(2P)^2(1+P)}$

1.87 = $\frac{(1+2P)^2}{(2P)^2(1+P)}$

① 0.2 $\frac{(1-0.2)^2}{(0.2)^2(1+0.2)}$

= $\frac{0.8 \times 0.8}{0.2 \times 0.2 \times 1.2}$

= 16

④ 0.4 $\frac{(1-0.4)^2}{(0.4)^2(1+0.2)}$

= $\frac{3.6 \times 0.6}{0.4 \times 0.4 \times 1.2}$

= 9

20.12.29
L-04

DALTON'S LAW OF PARTIAL PRESSURE :

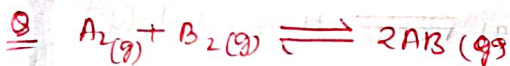
Mixture of gases, A, B, C.

$$P_A = (\text{mole fraction})_A \times \text{Total pressure}$$

$$P_A = \frac{n_A}{n_A + n_B + n_C} \cdot P_T$$

$$P_B = \frac{n_B}{n_A + n_B + n_C} \cdot P_T$$

$$P_C = \frac{n_C}{n_A + n_B + n_C} \cdot P_T$$



at eq. we have 0.2 moles of A_2 , 0.3 mole of B_2 and 0.1 moles of AB .

Total pressure is 6 atm. Find K_p .

① 0.666 ② 0.166 ③ 0.33 ④ 0.18.

Soln: $P_{A_2} = \frac{0.2}{0.2 + 0.3 + 0.1} \times 6 = \frac{0.2}{0.6} \times 6 = 2$

$P_{B_2} = \frac{0.3}{0.6} \times 6 = 3$

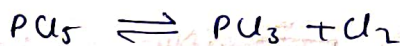
$P_{AB} = \frac{0.1}{0.6} \times 6 = 1$

$K_p = \frac{1^2}{2 \times 3} = \frac{1}{6} = 0.166$

DEGREE OF DISSOCIATION : α

The no. of moles of substance dissociated per mole of the substance.

$\rightarrow \alpha = 0.2$

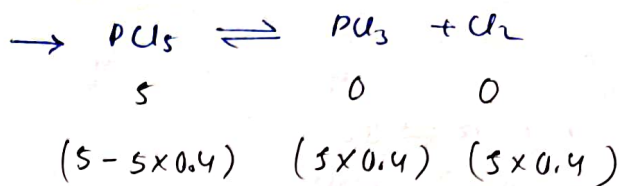
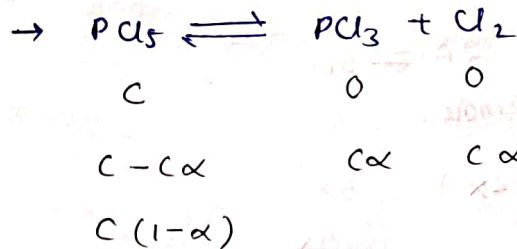


Initially 1 mole 0 0

At eq. $(1-\alpha)$ α α

$1-0.2$ 0.2 0.2

$= 0.8$ 0.2 0.2



$\alpha = 0.4$

Degree of dissociation = $\alpha \times 100$

$0 \leq \alpha \leq 1$

$\alpha = 0 \Rightarrow$ no dissociation

$\alpha = 1 \Rightarrow 100\%$ dissociation.

$\alpha = \frac{\text{no. of moles dissociated}}{\text{Total no. of moles.}}$ $\alpha = \frac{C\alpha}{C} = \alpha$

Q Find the value of α for each reactant for the following reactions-

TYPE-I:



1 mole

$(1-\alpha)$ (α)

$DOD = \frac{1-\alpha}{1}$

TYPE-II:



1 mole

$1-2\alpha$

or $1-\alpha$

α

$\alpha/2$

$DOD =$

TYPE-III:

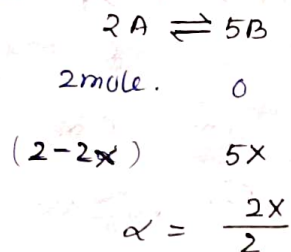


2 mole.

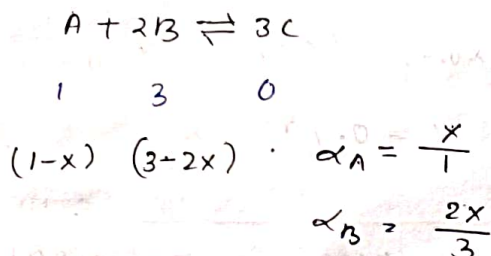
$2-2\alpha$

$\alpha = \frac{2\alpha}{2}$

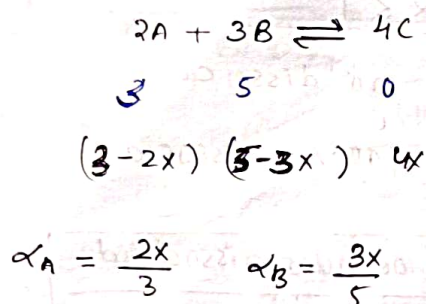
TYPE-IV :



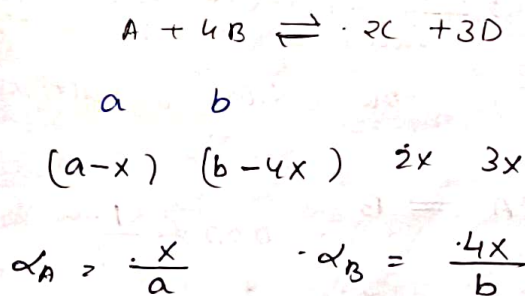
TYPE-V :



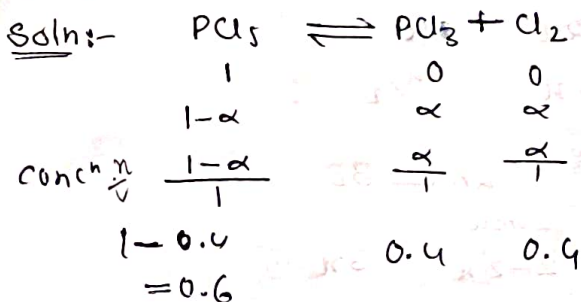
TYPE-VI :



TYPE-VII :



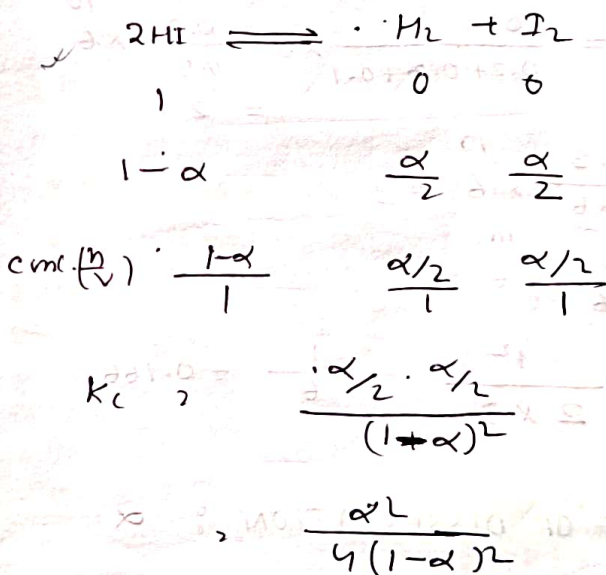
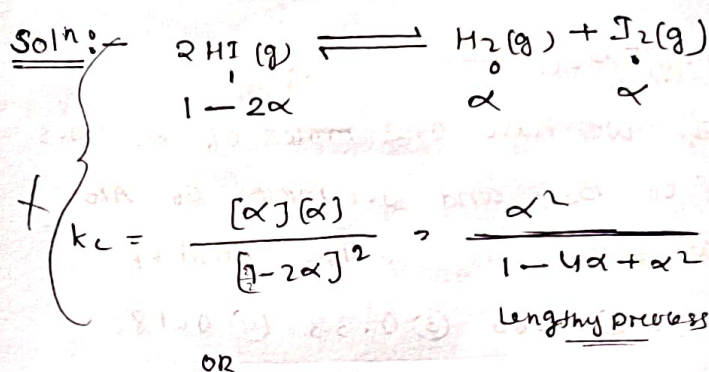
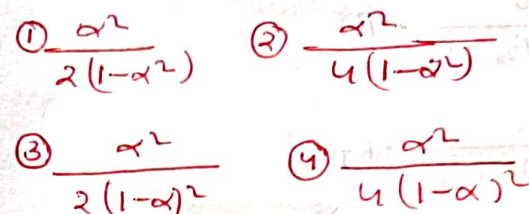
We have one mole of PCl_5 in 1L. vessel.
Find the concentration and eq. const. if $\alpha = 0.4$.



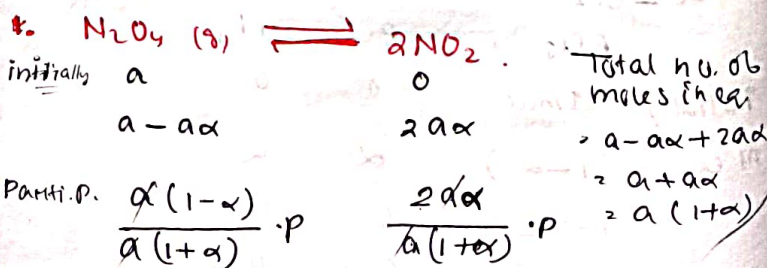
$$K_c = \frac{0.4 \times 0.4}{0.6} = \frac{0.16}{0.6} = \frac{8}{30} = 0.26$$



If DOD is α then find the eq. const. K_c



Q Write down the expression for K_p for the following reactions if initial concn is 'a' for the reactant.



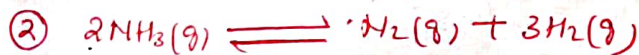
$$K_p = \frac{(P_{\text{NO}_2})^2}{(P_{\text{N}_2\text{O}_4})}$$

$$= \frac{\left(\frac{2\alpha}{1+\alpha} \cdot p\right)^2}{\left(\frac{1-\alpha}{1+\alpha} \cdot p\right)}$$

$$= \frac{2\alpha \cdot 2\alpha}{(1+\alpha)(1+\alpha)} \cdot p$$

$$= \frac{4\alpha^2}{(1+\alpha)(1-\alpha)} \cdot p$$

$$K_p = \frac{4\alpha^2}{1-\alpha^2} \cdot p$$



a	0	0	
$a - a\alpha$	$\frac{a\alpha}{2}$	$\frac{3a\alpha}{2}$	Total no of moles $= a - a\alpha + \frac{a\alpha}{2} + \frac{3a\alpha}{2}$
$a(1-\alpha)$			$= a - a\alpha + a\alpha$
mole. bractim: $\frac{a(1-\alpha)}{a(1+\alpha)}$	$\frac{a\alpha/2}{a(1+\alpha)}$	$\frac{3a\alpha/2}{a(1+\alpha)}$	$= a(1+\alpha)$
pp: $\frac{1-\alpha}{1+\alpha} \cdot p$	$\frac{\alpha}{2(1+\alpha)} \cdot p$	$\frac{3\alpha}{2(1+\alpha)} \cdot p$	

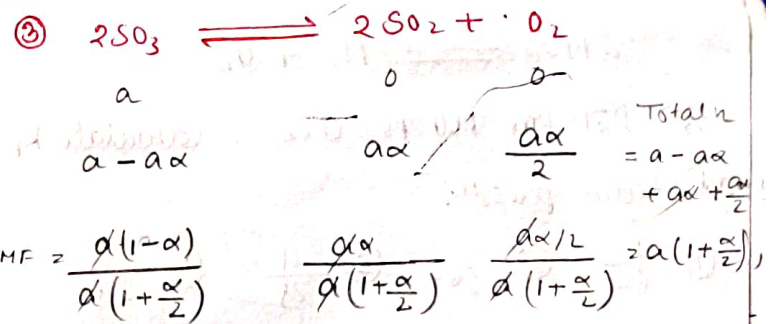
$$K_p = \frac{(P_{\text{N}_2})(P_{\text{H}_2})^3}{(P_{\text{NH}_3})^2}$$

$$= \frac{\frac{\alpha}{2(1+\alpha)} \cdot p \cdot \left(\frac{3\alpha}{2(1+\alpha)} \cdot p\right)^3}{\left(\frac{1-\alpha}{1+\alpha} \cdot p\right)^2}$$

$$= \frac{27\alpha^4 p^4}{16(1+\alpha)^4} \times \frac{(1+\alpha)^2}{(1-\alpha)^2} \cdot \frac{1}{p}$$

$$= \frac{27\alpha^4 p^2}{16(1+\alpha)(1+\alpha)(1-\alpha)(1-\alpha)}$$

$$= \frac{27\alpha^4 p^2}{16(1-\alpha^2)^2}$$



pp: $\frac{(1-\alpha)}{(1+\alpha/2)} \cdot p$ $\frac{\alpha}{(1+\alpha/2)} \cdot p$ $\frac{\alpha}{2(1+\alpha/2)} \cdot p$

$$K_p = \frac{(P_{\text{SO}_2})^2 (P_{\text{O}_2})}{(P_{\text{SO}_3})^2}$$

$$= \frac{\left(\frac{\alpha}{1+\alpha/2} \cdot p\right)^2 \left(\frac{\alpha}{2(1+\alpha/2)} \cdot p\right)}{\left(\frac{1-\alpha}{1+\alpha/2} \cdot p\right)^2}$$

$$= \frac{\frac{\alpha^2}{2(1+\alpha/2)} \cdot \frac{\alpha}{(1+\alpha/2)^2} \cdot p}{(1-\alpha)^2}$$

$$= \frac{\alpha^3 p}{2(1+\alpha/2)(1-\alpha)^2}$$

④ for the reaction,



If $\alpha = 0.25$, at 1 atmosphere pressure the value of K_p is close to.

① 0.25 ② 0.50 ③ 0.65 ④ 0.85

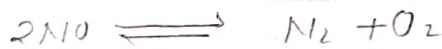
Soln: $\text{N}_2\text{O}_4 \rightleftharpoons 2\text{NO}_2$

1	0	
$1 - 0.25$ $= 0.75$	2×0.25 $= 0.5$	Total n $= 1 - 0.25 + 0.5$ $= 1.25$
MF: $\frac{0.75}{1.25}$	$\frac{0.5}{1.25}$	
pp: $\frac{3}{5} \times 1$	$\frac{2}{5} \times 1$	
$K_p = \frac{(P_{\text{NO}_2})^2}{(P_{\text{N}_2\text{O}_4})} = \frac{(2/5)^2}{3/5} = \frac{4}{15}$		

Q

Qb DOD of NO is 0.2. Calculate K_p at 3 atm pressure.

- ① $\frac{1}{64}$ ② $\frac{3}{64}$ ③ $\frac{1}{25}$ ④ $\frac{3}{25}$



$$\begin{array}{ccc} 1 & 0 & 0 \\ 1-0.2 & 0.1 & 0.1 \\ = 0.8 & & \end{array}$$

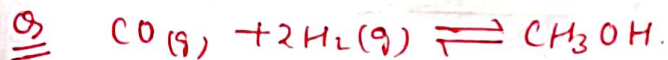
$$M_t \quad \frac{0.8}{1} \quad \frac{0.1}{1} \quad \frac{0.1}{1}$$

$$P_P = \frac{0.8}{1} \times 3 \quad \frac{0.1}{1} \times 3 \quad \frac{0.1}{1} \times 3$$

$$K_p = \frac{0.3 \times 0.3}{(2.4)^2}$$

$$\frac{0.3 \times 0.3}{2.4 \times 2.4}$$

$$= \frac{1}{64}$$



0.2 moles of CO in a sealed flask is made to react with H_2 at 327°C (600 K). at eq. 0.1 mole of CH_3OH are formed.

Qb K_p for reaction is -

$$K_p = \frac{\alpha}{(4.92)^2} \quad \text{Calculate } \alpha \text{ if}$$

pressure at eq. is 4.92 atm.

- ① $\frac{9}{16}$ ② $\frac{9}{25}$ ③ $\frac{16}{9}$ ④ $\frac{25}{9}$

Soln:-

Soln:



$$0.2 \quad a \quad 0$$

$$0.2-x \quad a-2x$$

$$0.2-0.1 \quad a-2 \times 0.1$$

$$= 0.1$$

$$a-0.2$$

$$a=0.5$$

$$0.5-0.2=0.3$$

$$V = 8\text{L}$$

$$T = 600\text{K}$$

$$P = 4.92$$

$$PV = nRT$$

$$n = \frac{PV}{RT} = \frac{4.92 \times 8}{1 \times 600} = 0.5 = a$$

$$P_{\text{CH}_3\text{OH}} = \frac{0.1}{0.5} \times P$$

$$P_{\text{H}_2} = \frac{0.3}{0.5} \times P$$

$$P_{\text{CO}} = \frac{0.1}{0.5} \times P$$

$$K_p = \frac{P_{\text{CH}_3\text{OH}}}{P_{\text{CO}} \cdot (P_{\text{H}_2})^2}$$

$$= \frac{0.1/0.5 \times P}{0.1/0.5 \times P}$$

$$\frac{0.3}{0.5} \times \frac{0.3}{0.5} \times P^2 \times \frac{0.1}{0.5} \times P$$

$$= \frac{0.25}{0.9 \times 0.1} = \frac{25}{9} / (4.92)^2$$

$$\frac{K_p \alpha}{(4.92)^2} = \frac{25}{9} / (4.92)^2$$

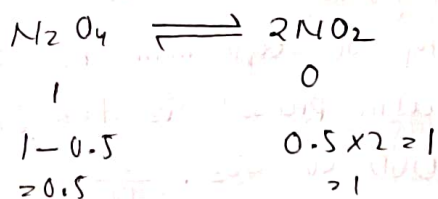
$$\therefore \alpha = \frac{25}{9}$$

22.12.24

L-06

Q 96 50% of N_2O_4 is dissociated at 2 atm pressure. then calculate the K_p for the rxn.

Soln



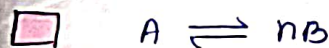
$$\text{PP} = \frac{0.5}{1.5} \times 2 \quad \frac{1}{1.5} \times 4 \quad \left\{ n = 1.5 \right.$$

$$= \frac{2}{3} \quad \frac{4}{3}$$

$$K_p = \frac{\left(\frac{4}{3}\right)^2}{\frac{2}{3}} = \frac{\frac{4}{3} \times \frac{4}{3}}{\frac{2}{3}} = \frac{8}{3} = 2.66$$

⊖ AVERAGE MOLAR MASS / OBSERVED

MOLAR MASS / VAPOUR DENSITY / DOD :



Initial: 1 0

At eq: $1-\alpha$ $n\alpha$

$$\begin{aligned} \text{Total no. of moles} &= 1-\alpha + n\alpha \\ &= 1+\alpha(n-1) \end{aligned}$$

$$M_{\text{obs}} = \frac{\text{Total mass}}{\text{Total moles}}$$

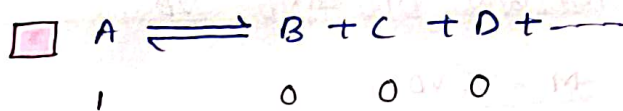
$$\boxed{M_{\text{obs}} = \frac{M_A}{1+\alpha(n-1)}} \quad \begin{array}{l} n = \text{no. of moles of} \\ \text{pdt obtained by} \\ 1 \text{ mole of reactant.} \end{array}$$

$$n = \frac{\text{given wt}}{\text{M.W.}}$$

$$\text{Molar mass} = \frac{\text{Molar mass}}{\text{no. of moles}}$$

$$\boxed{M_{\text{obs}} = \frac{\text{Total mass}}{\text{Total no. of moles}}}$$

$$M_{\text{obs}} = \frac{M_{\text{each}}}{\text{Total no. of moles}}$$



1 0 0 0 ...

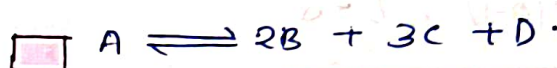
$1-\alpha$ α α α ...

$$\text{Total no. of moles} = 1-\alpha + \alpha + \alpha + \alpha + \dots$$

$$= 1-\alpha + n\alpha$$

$$= 1+\alpha(n-1)$$

$$M_{\text{obs}} = \frac{M}{1+\alpha(n-1)}$$



1 0 0 0

$1-\alpha$ 2α 3α α

$$n = 6$$

⊖ For the M_xN_y



The observed molar mass of N_2O_4 at eqm is 80 then calculate α .

$$\Rightarrow \text{N}_2\text{O}_4; \text{MW} = 28 \times 2 = 56$$

$$M_{\text{obs}} = \frac{M_{\text{each}}}{\text{Total no. of moles}}$$

$$80 = \frac{56}{1+\alpha(2-1)}$$

$$80 = \frac{56}{1+\alpha(1)}$$

$$80 + 80\alpha = 56$$

$$80\alpha = 56 - 80$$

$$\alpha = \frac{-24}{80}$$

$$= 0.3$$

RELATION B/W DOD AND VD :

$$M = 2VD.$$

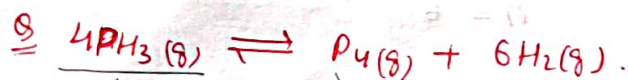
$$2(VD)_{\text{after dissociation}} = \frac{2(VD)_{\text{before dissociation}}}{1 + \alpha(n-1)}$$

$$d = \frac{D}{1 + \alpha(n-1)} \quad \begin{matrix} d = VD \text{ after diss.} \\ D = VD \text{ before diss.} \end{matrix}$$

$$d + d\alpha(n-1) = D$$

$$d \cdot \alpha(n-1) = D - d$$

$$\alpha = \frac{D - d}{d(n-1)}$$



9b. Obs. VD is 13. Then calculate the DOD of PH_3 (phosphine).

Soln:- $VD = \frac{MW}{2} = \frac{34}{2} = 17.$

$$\alpha = \frac{D - d}{d(n-1)} = \frac{17 - 13}{13(7/4 - 1)}$$

$$4 \rightarrow 7$$

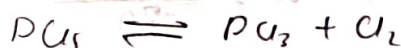
$$1 \rightarrow 7/4$$

$$= \frac{42}{13(7/4 - 1)}$$

$$= \frac{4}{13 \cdot \frac{3}{4}}$$

$$= \frac{16}{39}$$

8 The obs VD of PCl_5 at eqm is 62. 9b VD of PCl_5 initially is 104 calculate the DOD.



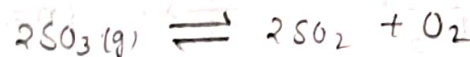
$$n = 2$$

$$\alpha = \frac{104 - 62}{62(2-1)} = \frac{42}{62} \times 100 = 67.2\%$$



Density of eqm mixture at 727 and 2 atm pressure is 1.7 g/l. Calculate DOD of SO_3 . $\frac{6}{17}.$

Soln:-



$$n = \frac{3}{2}$$

$$M_{\text{obs}} = \frac{M_r}{1 + (n-1)\alpha} = \frac{80}{1 + (\frac{3}{2} - 1)\alpha}$$

$$M = \frac{80}{1 + \alpha/2}$$

$PM = dRT$

$$d = \frac{PM}{RT}$$

$$1.7 = \frac{2 \times \frac{80}{1 + \alpha/2}}{0.08 \times 1000}$$

$$1.7 = \frac{2 \times \frac{80}{1 + \alpha/2}}{80}$$

$$1 + \alpha/2 = \frac{0.2}{1.7}$$

$$\alpha/2 = \frac{2}{1.7} - 1$$

$$= \frac{2 - 1.7}{1.7} = \frac{0.3}{1.7}$$

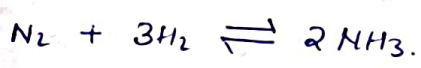
$$\alpha = \frac{2 \times 0.3}{1.7} = \frac{0.6}{1.7} = \frac{6}{17}$$

LE - CHATELIER'S PRINCIPLE :

- This principle is only applicable for reversible process.
- Eqm state of any reversible reaction is the most stable state and if any change is made by changing pressure, volume, temperature or addition of inert material then eqm of the reaction will be disturbed and reaction will move in that direction from where eqm can exist again and effect of applied change is nullified.
Punching bag → Manna ke baad bhi apne jagan aa jata hai.
Yani ki apni eqm state mein.

1. CHANGE IN CONCⁿ OR MOLES :

- On increasing concⁿ of reactant, reaction will move in forward direction and on increase in concⁿ of product, rxn will move in backward direction.



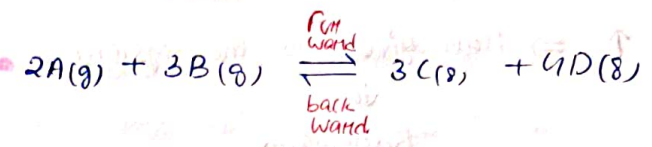
$[N_2] \quad [H_2] \quad [NH_3] \rightarrow$ all are in eqm.

- concⁿ of N_2 : $[N_2] \uparrow \rightarrow$ forward.
- concⁿ of H_2 : $[H_2] \uparrow \rightarrow$ forward.
- $[NH_3] \uparrow \rightarrow$ backward.
- Removal of product → best method.

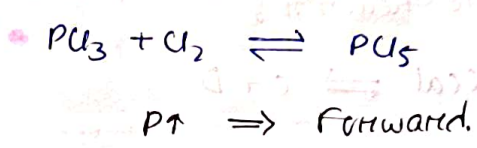
• most suitable condition in terms of concⁿ to res the yield of the pdt for any rxn is by incⁿ of reactant and continuous removal of product.

2. CHANGE IN PRESSURE :

• If we increase the pressure of any reaction then eqm will shift to that direction where less no. of gaseous species are present.



$P \uparrow \Rightarrow$ Less no. of moles
 $\Downarrow$
 backward.



Reason :

① $P \uparrow = V \downarrow \quad P \propto \frac{1}{V}$

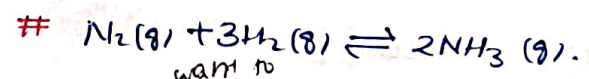
$$C_{mole} = \left[\frac{\text{moles}}{V} \right] \downarrow$$

concⁿ $\uparrow$

$P \uparrow \Rightarrow$ Eqm will $\rightarrow$ that direction where less no. of moles.

② $PV = nRT$
 $P \propto n$

$P \uparrow \Rightarrow n \uparrow \rightarrow$ Eqm will decrease the no. of moles.



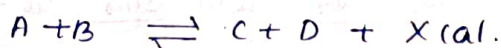
→ If we increase the production of NH_3
 $\Rightarrow P \uparrow$

NOTE :

- ① $\Delta n_g = 0$, no effect of P.
- ② $\Delta n_g > 0$ or $\Delta n_g < 0$, there will be change in P.
- ③ For solid and liquid, no effect of P.

3. CHANGE IN TEMPERATURE :

• For exothermic reaction:



$$\Delta H = -x \text{ cal}$$

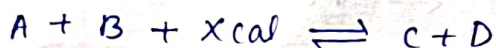
Temp $\uparrow \Rightarrow$ Heat given to the system

$\Downarrow$

Backward

Temp $\downarrow \Rightarrow$ Forward.

• For endothermic reaction:



$$\Delta H = +x \text{ cal.}$$

Temp $\uparrow \Rightarrow$ Forward.

Temp $\downarrow \Rightarrow$ backward.

4. ADDITION OF INERT GAS :

(added inert gas should not react with the eqm mixture)

• The addition of inert gas can define in two ways.

① At const. volume:

$\rightarrow$ no effect.

② At const pressure:

$\rightarrow$ eqm will shift towards that side where more no. of moles are present.

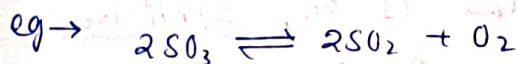
• At const. P. addition of inert gas.

$\Downarrow$

Volm $\uparrow$

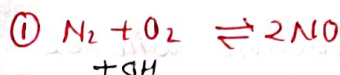
$$\left[\text{cmcn} = \frac{n}{\text{Volm}} \right]$$

cmcn $\downarrow \Rightarrow$ Towards more no. of moles.



inert gas $\Rightarrow$ forward.

③ Predict the conditions how can we achieve the max yield of the pdt. for the following reaction.

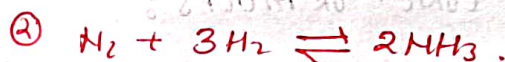


① $[\text{N}_2] \uparrow \rightarrow$ forward.

② $[\text{O}_2] \uparrow$

③ Removal of (NO) product.

④ Temp $\uparrow$



① $[\text{N}_2] \uparrow$

② $[\text{H}_2] \uparrow$

③ Product (NH_3) should be removed.

④ Pressure $\uparrow$

⑤ Temp $\uparrow$

⑤ REACTION QUOTIENT :



$$Q = \frac{[\text{Pdt}]}{[\text{Reactant}]}$$

$$Q = \frac{[C]^{m_1} [D]^{m_2}}{[A]^{n_1} [B]^{n_2}} \rightarrow \text{At any cmcn or at any time of the reaction}$$

$$K_{eq} = \frac{[C]^{m_1} [D]^{m_2}}{[A]^{n_1} [B]^{n_2}} \rightarrow \text{At eqm cmcn}$$

Condition:

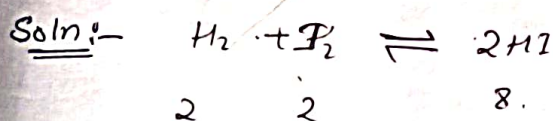
- ① $Q = K_{eq}$, at equilibrium
- ② $Q > K_{eq}$, backward reaction
- ③ $Q < K_{eq}$, forward reaction.

Q. What will be the direction of rxn if concn H_2 , I_2 , and HI are 2 mol/l , 2 mol/l , 8 mol/l respectively.

K_{eq} for the rxn $H_2 + I_2 \rightleftharpoons 2HI$ is

4.

- ① forward direction.
- ② backward direction.
- ③ Equilibrium condition.
- ④ Reaction will be completed.



$$Q = \frac{8^2}{2 \times 2} = 16$$

$Q > K_{eq}$, backward.

HW $\Rightarrow$ B3-S, 6

Ex-1 $\Rightarrow$ complete.

24.12.24
L-08

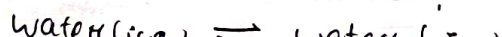
PHYSICAL EQUILIBRIUM:

PHYSICAL REACTION:

Those reaction in which change in only and only physical states (solid, liq and gas) of substance takes place without any chemical change is called physical reaction.

eg $\rightarrow$ ① Ice-water system (melting

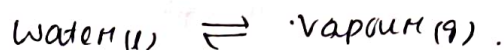
$\rightarrow$ Melting of ice is accompanied by absorption of heat (endothermic) and decrease in volume.



Hence both increase in temp and pressure will favour the melting of ice into water.

② Water-water vapour system: (vapourisation of water)

Vapourisation of water is an endothermic process and condensation of vapour into water is an exothermic rxn.

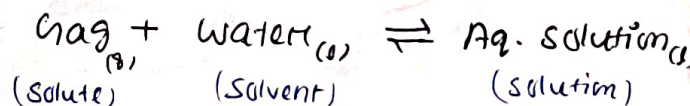


When the temp is increased, so, rise in temp will increase the vapour.

The eqm shift towards left hand side when the pressure is increased (that is volume is decreased) - so increase in pressure will favour the rate of condensation of vapour into water.

The favourable conditions for conversion water into vapour are high temp and low pressure.

③ Solubility of gases:



Effect of pressure: Solubility of such gases increases with increase in pressure which dissolves in a solvent with a decrease in volume.